

QF1100 Introduction to Quantitative Finance

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Contents

1	Theory of Interest	2
2	Bond & Term Structure	3
3	Asset Returns and Probability	4
3.1	Asset Returns	4
3.1.1	Random Return of Assets	5
3.2	Portfolio Returns	6
3.2.1	Portfolio of Two Assets	6
3.2.2	Portfolio of n Assets	6
3.2.3	Mean-Variance Analysis	7
3.2.4	Feasible Sets	8
4	Forwards & Futures	9
5	Options	10
6	Cheatsheet	11

1 Theory of Interest

2 Bond & Term Structure

3 Asset Returns and Probability

3.1 Asset Returns

Motivation

Suppose an investor **purchases** an asset at time $t = 0$, and sells it a year later at time $t = 1$. Let P_i denote the price at time $t = i$ (in years). Then, we define the **total return** to be

$$R = \frac{\text{amt received}}{\text{amt invested}} = \frac{P_1}{P_0},$$

and the **rate of return** to be

$$r = \frac{\text{amt received} - \text{amt invested}}{\text{amt invested}} = \frac{P_1 - P_0}{P_0}.$$

We interpret R as the amount gained *per dollar* invested. For example, if we buy gold for \$50 and sell it for \$60, then $R = 60/50 = 1.20$, meaning we earned \$1.20 for every \$1 spent. If $R > 1$ we earned money; if $R < 1$ we lost money. We reconcile this with r via percentages: in the example above we gained a 20% return, so $r = 0.20$.

Notice that $R = 1 + r$, and in particular $P_1 = (1 + r)P_0$, so r acts as the interest gained.

Next, we introduce the notion of **longing** and **shorting**.

Definition 3.1 (Short-selling). The process of **short-selling** (or *shorting*) an asset proceeds as follows.

1. *Borrow* the asset from someone who owns it.
2. *Sell* the borrowed asset immediately to receive P_0 .
3. At a later date, anticipating a drop in price, *repurchase* the asset for a lower price P_1 , turning a profit of $P_0 - P_1$, then return it to the lender.

Random Variables

We recall standard probability terminology.

Definition 3.2 (Expected value). $\mathbb{E}(X) = \sum x p(x)$. Key properties:

1. $\mathbb{E}(c) = c$
2. $\mathbb{E}(cX) = c \mathbb{E}(X)$
3. $\mathbb{E}(X + c) = \mathbb{E}(X) + c$
4. $\mathbb{E}(aX + bY) = a \mathbb{E}(X) + b \mathbb{E}(Y)$

Definition 3.3 (Variance, standard deviation, covariance, correlation). 1. $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}(X))^2]$

2. $\text{SD}(X) = \sqrt{\text{Var}(X)}$
3. $\sigma_{X,Y} = \text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$
4. $\rho_{X,Y} = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}, \quad -1 \leq \rho_{X,Y} \leq 1$

Lemma 3.4 (Properties of covariance). 1. $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$

2. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
3. $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$
4. $\text{Var}(X + Y) = \text{Var}(X) + 2 \text{Cov}(X, Y) + \text{Var}(Y)$

3.1.1 Random Return of Assets

At time $t = 0$, we do not know P_1 . We therefore treat it as a random variable, making R and r random variables as well.

Definition 3.5 (Mean, variance, and covariance of returns). Given asset k with rate of return r_k , we define:

- **Mean:** $\mu_k = \mathbb{E}(r_k)$
- **Variance:** $\sigma_k^2 = \text{Var}(r_k)$
- **Covariance** of assets i, j : $\sigma_{ij} = \text{Cov}(r_i, r_j)$
- **Correlation coefficient:** $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}$

3.2 Portfolio Returns

Motivation

Now, instead of a single asset, we consider a portfolio of n assets. Let X_0 denote the total amount invested. We choose weights w_i such that $X_{0,i} = w_i X_0$ is the amount invested in the i -th asset, satisfying

$$\sum_{i=1}^n X_{0,i} = X_0 \quad \Longleftrightarrow \quad \sum_{i=1}^n w_i = 1.$$

Remark 3.6. If short-selling is allowed, we permit $w_i < 0$. Otherwise we require $w_i \geq 0$ for all i .

3.2.1 Portfolio of Two Assets

For the case of *two* assets, our weights will of course be

$$w_1 = \alpha \quad w_2 = 1 - \alpha$$

Proposition 3.7 (Returns for a portfolio of 2 assets). *Given a portfolio of two assets, the wealth at the end of the period is*

$$\begin{aligned} X_1 &= \alpha X_0(1 + r_1) + (1 - \alpha)X_0(1 + r_2) \\ &= X_0 \left[1 + \left\{ \alpha r_1 + (1 - \alpha)r_2 \right\} \right] \end{aligned}$$

So, the rate of return is $\alpha r_1 + (1 - \alpha)r_2$.

3.2.2 Portfolio of n Assets

We can then generalise the returns in the case of 2 assets to the case of n assets.

Proposition 3.8. *Given a portfolio of n assets, the wealth at the end of the period is*

$$X_1 = \sum_{i=1}^n w_i X_0(1 + r_i) = X_0 \sum_{i=1}^n w_i(1 + r_i)$$

So, the rate of return is $r_p = \sum_{i=1}^n w_i r_i$.

Proof. Let $X_0(1 + r_p) = X_0 \sum_{i=1}^n w_i(1 + r_i)$. Then by comparing terms,

$$\begin{aligned} r_p &= \left(\sum_{i=1}^n w_i(1 + r_i) \right) - 1 \\ &= \sum_{i=1}^n w_i + \left(\sum_{i=1}^n w_i r_i \right) - 1 \\ &= \sum_{i=1}^n w_i r_i \quad \text{bc. } \sum_{i=1}^n w_i = 1 \end{aligned}$$

□

Now, we also want to find the portfolio *mean* and *variance*.

Proposition 3.9 (Portfolio Mean). *The mean rate of return of a portfolio of n assets is given by*

$$\mu_p = \mathbb{E}(r_p) = \sum_{i=1}^n w_i \mu_i$$

where $\mu_i = \mathbb{E}(r_i)$.

Proof. Follows from linearity of expectation. □

Proposition 3.10 (Portfolio Variance). *The variance of the rate of return of a portfolio of n assets is given by*

$$\sigma_p^2 = \text{Var}(r_p) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij}$$

where $\sigma_{ij} = \text{Cov}(r_i, r_j)$

Proof. Use definition of variance to derive. □

3.2.3 Mean-Variance Analysis

The **mean-variance analysis** aims to ask the question: *As α varies, how will μ_p and σ_p change?*

We first recall the portfolio variance of n variables (3.10) and look at the case of two assets.

Proposition 3.11. *The variance of the portfolio return is given by*

$$\begin{aligned} \sigma_p^2 &= \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \sigma_{12} \\ &= \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \rho_{12} \sigma_1 \sigma_2 \end{aligned}$$

Proof. We use the following result:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

So,

$$\begin{aligned} \sigma_p^2 &= \text{Var}(r_p) \\ &= \text{Var}(\alpha r_1 + (1 - \alpha) r_2) \\ &= \alpha^2 \text{Var}(r_1) + (1 - \alpha)^2 \text{Var}(r_2) + 2 \text{Cov}(\alpha r_1, (1 - \alpha) r_2) \\ &= \alpha^2 \text{Var}(r_1) + (1 - \alpha)^2 \text{Var}(r_2) + 2\alpha(1 - \alpha) \text{Cov}(r_1, r_2) \\ &= \alpha^2 \sigma_1^2 + (1 - \alpha)^2 \sigma_2^2 + 2\alpha(1 - \alpha) \sigma_{12}, \end{aligned}$$

where we used the bilinearity of covariance to pull out the scalar factors α and $(1 - \alpha)$, and set $\sigma_{12} = \text{Cov}(r_1, r_2)$. The second line follows from the definition $\sigma_{12} = \rho_{12} \sigma_1 \sigma_2$, where ρ_{12} is the correlation coefficient between r_1 and r_2 . □

And in *mean-variance analysis*, we are interested to find the α that minimises this σ_p^2 , that is,

$$\alpha^* = \arg \min_{\alpha \in \mathbb{R}} \sigma_p^2$$

Proposition 3.12. *The minimum portfolio variance σ_p^2 occurs when*

$$\alpha^* = \frac{\sigma_2(\sigma_2 - \rho_{12}\sigma_1)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

*and the minimum portfolio **variance** and **mean** is given by*

$$(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1 - \rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$$

$$(\mu_p)^* = \alpha^*\mu_1 + (1 - \alpha^*)\mu_2$$

This is called the **global minimum-variance portfolio (GMVP)**.

3.2.4 Feasible Sets

4 Forwards & Futures

5 Options

6 Cheatsheet
